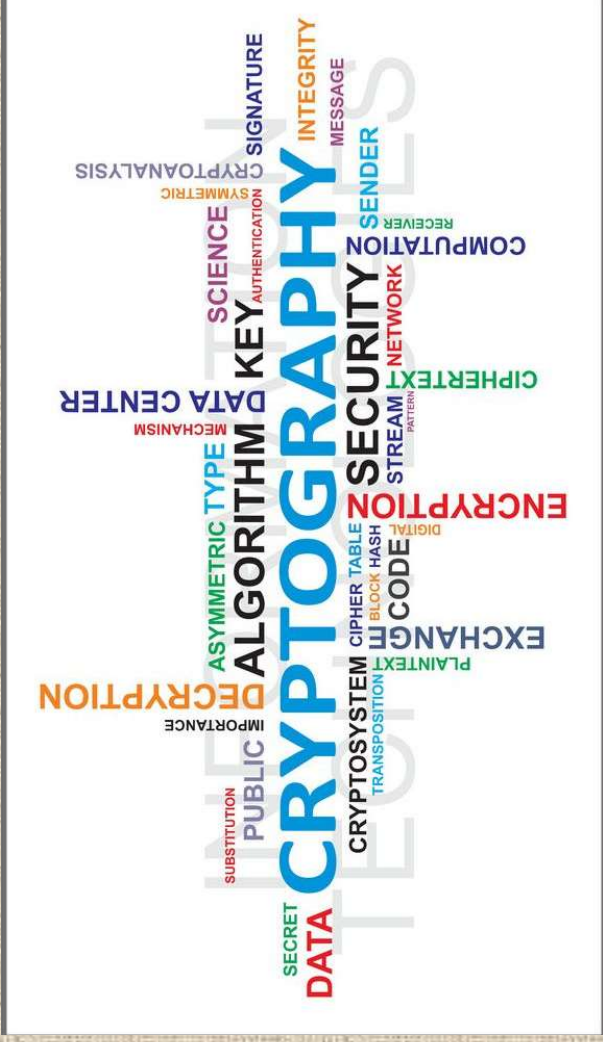


Asymmetric Key Cryptography

Public Key Cryptography



Public-Key Cryptography

Public Key cryptography is Also known as asymmetric-key cryptography.

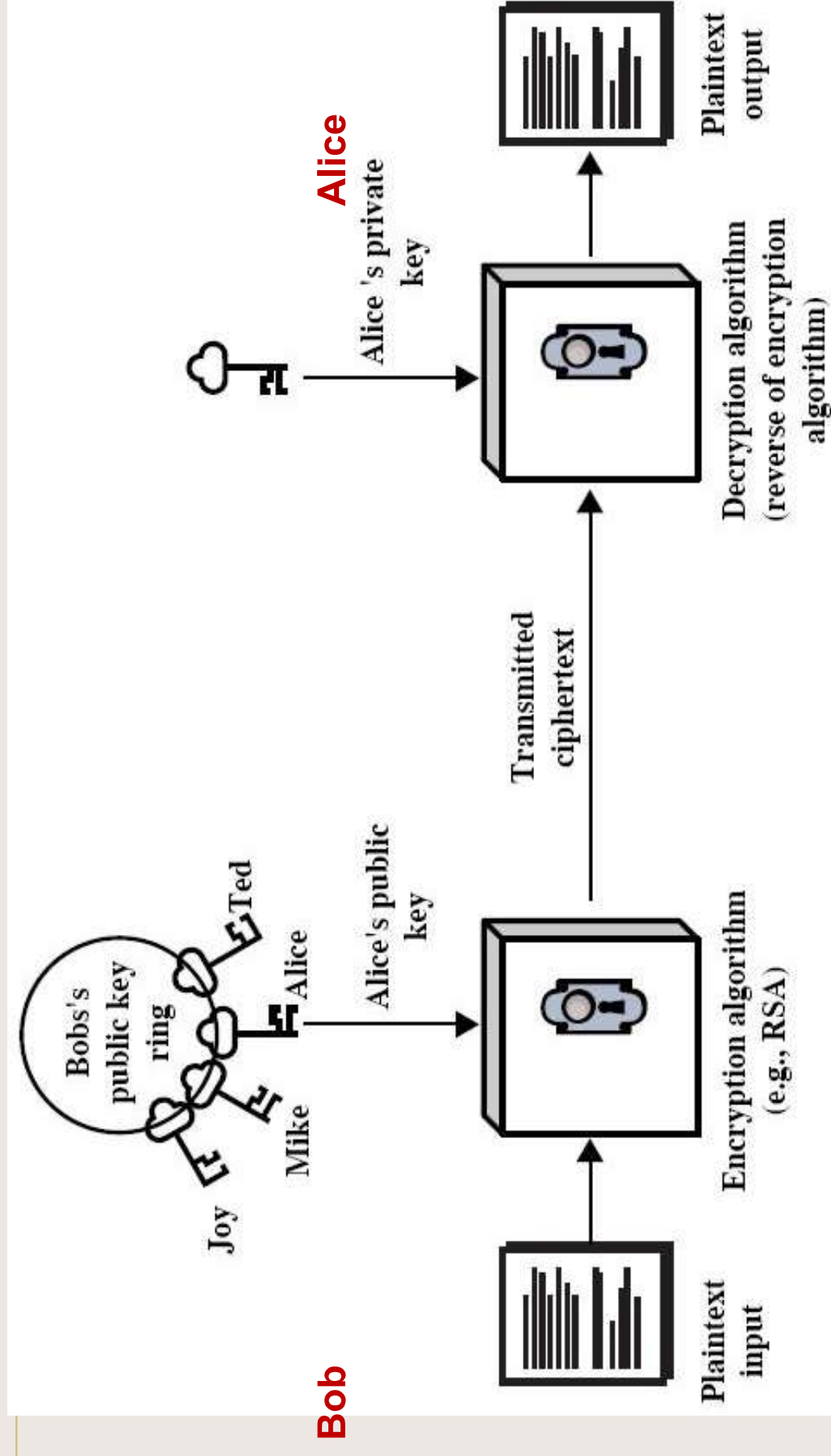
A pair of keys : a **public key** and a **private key** used for encryption of plaintext and decryption of cipher text respectively. Examples are RSA, Diffie-Hellman, Elgamal, ECC, ECC-Diffe-Hellman.

- The public key is used for encryption and verify signatures .
- The key is known to the public.
- The private key is used for decryption sign (create) signatures.
- The key is only known to the owner. The private key is kept secret.
- Public-key algorithms are based on mathematical functions rather than on substitution and permutation.
- Security of any encryption scheme is depend on the length of the key and the computational work.

Why Public-Key Cryptography?

- Developed to address two key issues:
 - **Key distribution** – how to have secure communications in general without having to trust a Key Distributed Center.
 - **Digital signatures** – how to verify a message comes intact from the claimed sender
 - Invented by Whitfield Diffie & Martin Hellman in 1976.

Public-Key Cryptography



Public-Key Cryptosystems: Secrecy

A that produces a message in plaintext,

$$X = [X_1, X_2, \dots, X_M].$$

The message is intended for destination B. B generates a related pair of keys: a **public key**, PU_b , and a **private key**, PR_b . PR_b is known only to B, whereas PU_b is publicly available and therefore accessible by A.

With the message X and the encryption key PU_b as input, A forms the ciphertext

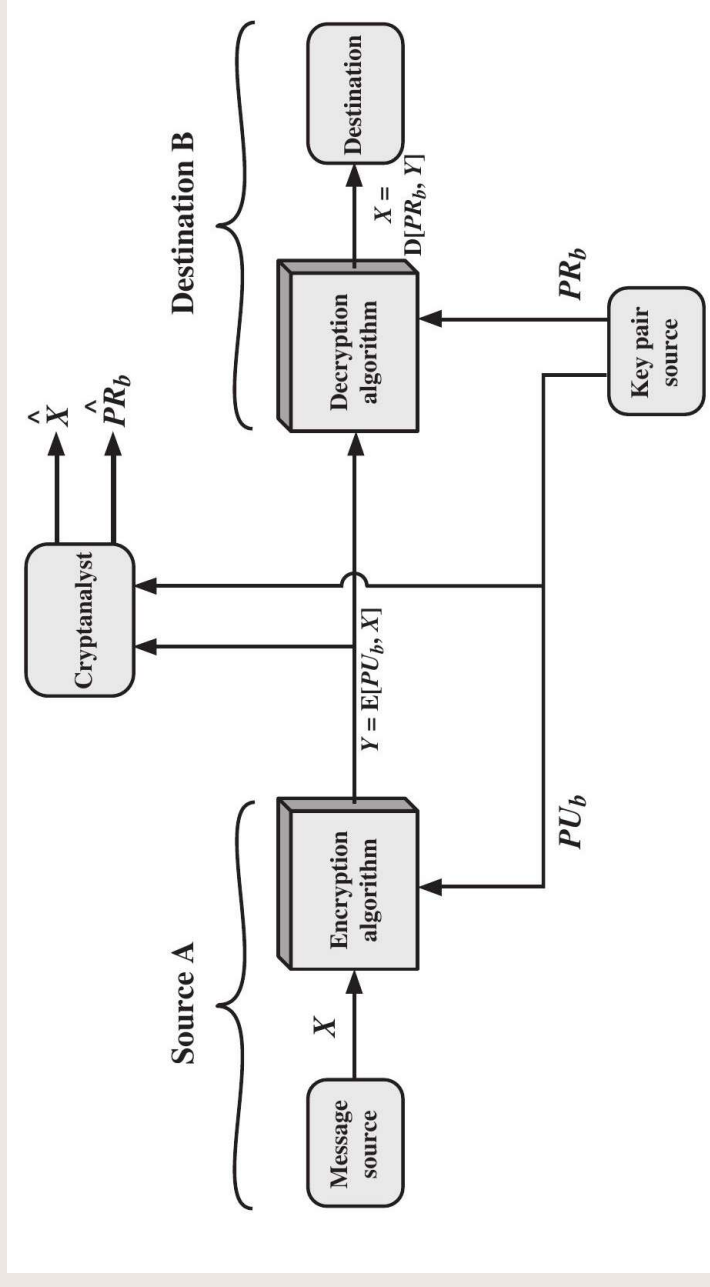
$$Y = [Y_1, Y_2, \dots, Y_N]:$$

$$Y = E(PU_b, X)$$

The intended receiver, in possession of the matching private key, is able to invert the transformation:

$$X = D(PR_b, Y)$$

provides confidentiality



Public-Key Cryptosystem: Authentication

A prepares a message to B and encrypts it using A's private key before transmitting it.

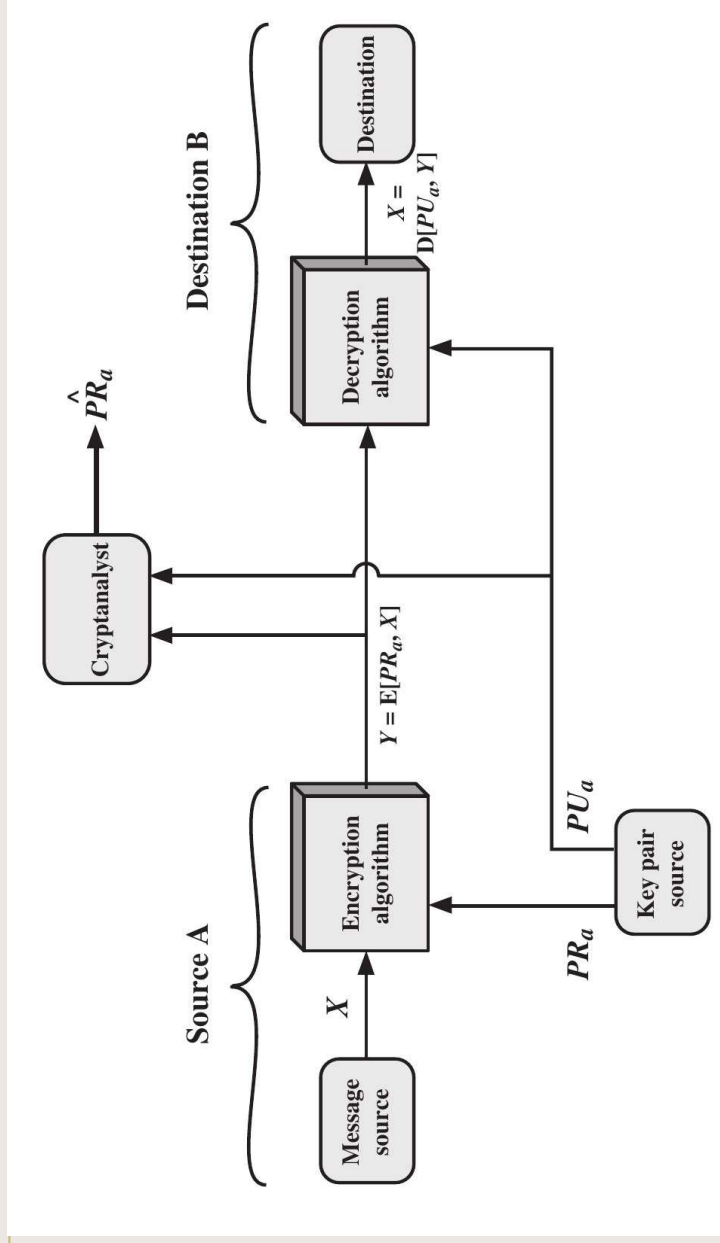
$$Y = E(PR_a, X)$$

B can decrypt the message using A's public key.

$$X = D(PU_a, Y)$$

Because the message was encrypted using A's private key, only A could have prepared the message. Therefore, the entire encrypted message serves as a digital signature.

In addition, it is impossible to alter the message without access to A's private key, so the message is authenticated both in terms of source and in terms of data integrity.



Public-Key Applications

- Can classify uses into 3 categories:
 - **Encryption/decryption (provide secrecy)** : The sender encrypts a message with the recipient's public key.
 - **Digital signature (provide authentication)** : The sender "signs" a message with its private key, either to the whole message or to a small block of data that is a function of the message.
 - **Key exchange (session keys)** : Two sides cooperate to exchange a session key. Several different approaches are possible, involving the private key(s) of one or both parties.

RSA (Rivest-Shamir-Adleman) Algorithm

- Each user generates a public/private key pair by selecting two large primes at random - p, q
- Computing their system modulus $n=p \cdot q$
 - note $\phi(n)=(p-1)(q-1)$
- Selecting at random the encryption key e
 - where $1 < e < \phi(n)$, $\gcd(\phi(n), e) = 1$
- Solve following equation to find decryption key d
 - $e \cdot d \bmod \phi(n) = 1$ and $0 \leq d \leq n$
- Publish their public encryption key: $PU = \{e, n\}$
- Keep secret private decryption key: $PR = \{d, n\}$

RSA Encryption/decryption

- to encrypt a message M the sender:
 - obtains **public key** of recipient $PU = \{e, n\}$
 - computes: $C = M^e \bmod n$, where $0 \leq M < n$
- to decrypt the ciphertext C the owner:
 - uses their **private key** $PR = \{d, n\}$
 - computes: $M = C^d \bmod n$
- note that the message M must be smaller than the modulus n (block if needed)

Why RSA Works

- It is infeasible to determine d given e and n
- because of Euler's Theorem:
 - $a^{\phi(n)} \bmod n = 1$ where $\gcd(a, n) = 1$
- in RSA have:
 - $n = p \cdot q$
 - $\phi(n) = (p-1)(q-1)$
 - carefully choose e & d to be inverses $\bmod \phi(n) \Rightarrow ed \bmod \phi(n) = 1$

RSA Example - Key Setup

1. Select primes: $p=17$ & $q=11$
2. Calculate $n = pq = 17 \times 11 = 187$
3. Calculate $\phi(n) = (p-1)(q-1) = 16 \times 10 = 160$
4. Select e between 1 and 160:
 $\gcd(e, 160) = 1$; choose $e=7$
5. Determine d : $de \bmod 160 = 1$ and $d < 160$
Value is $d=23$ since $23 \times 7 = 161$
6. Publish public key $PU = \{7, 187\}$
7. Keep secret private key $PR = \{23, 187\}$

RSA Example - En/Decryption

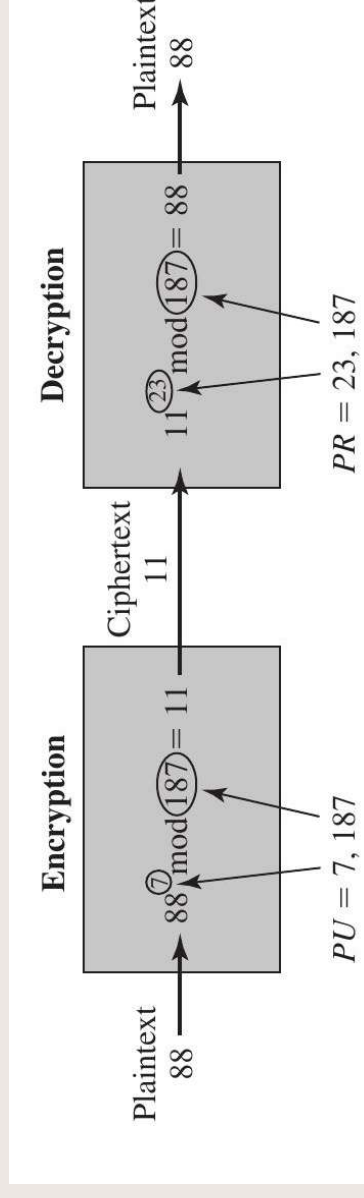
- sample RSA encryption/decryption is:
- given message $M = 88$ (nb. $88 < 187$)

- encryption:

$$C = 88^7 \bmod 187 = 11$$

- decryption:

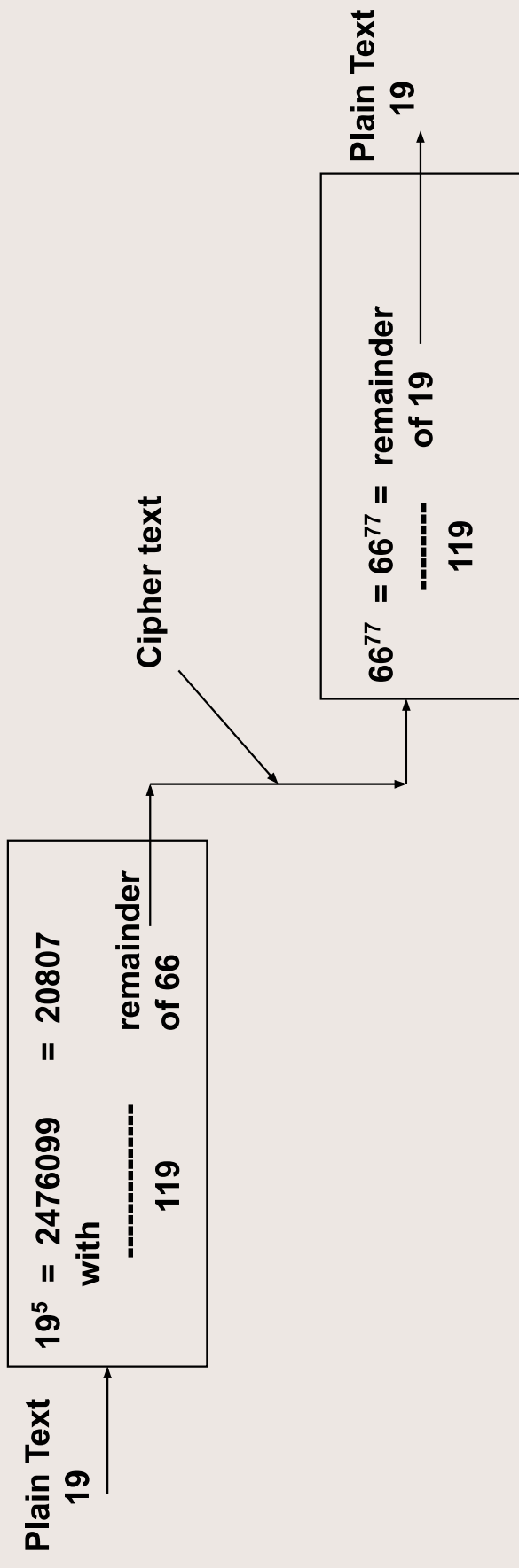
$$M = 11^{23} \bmod 187 = 88$$



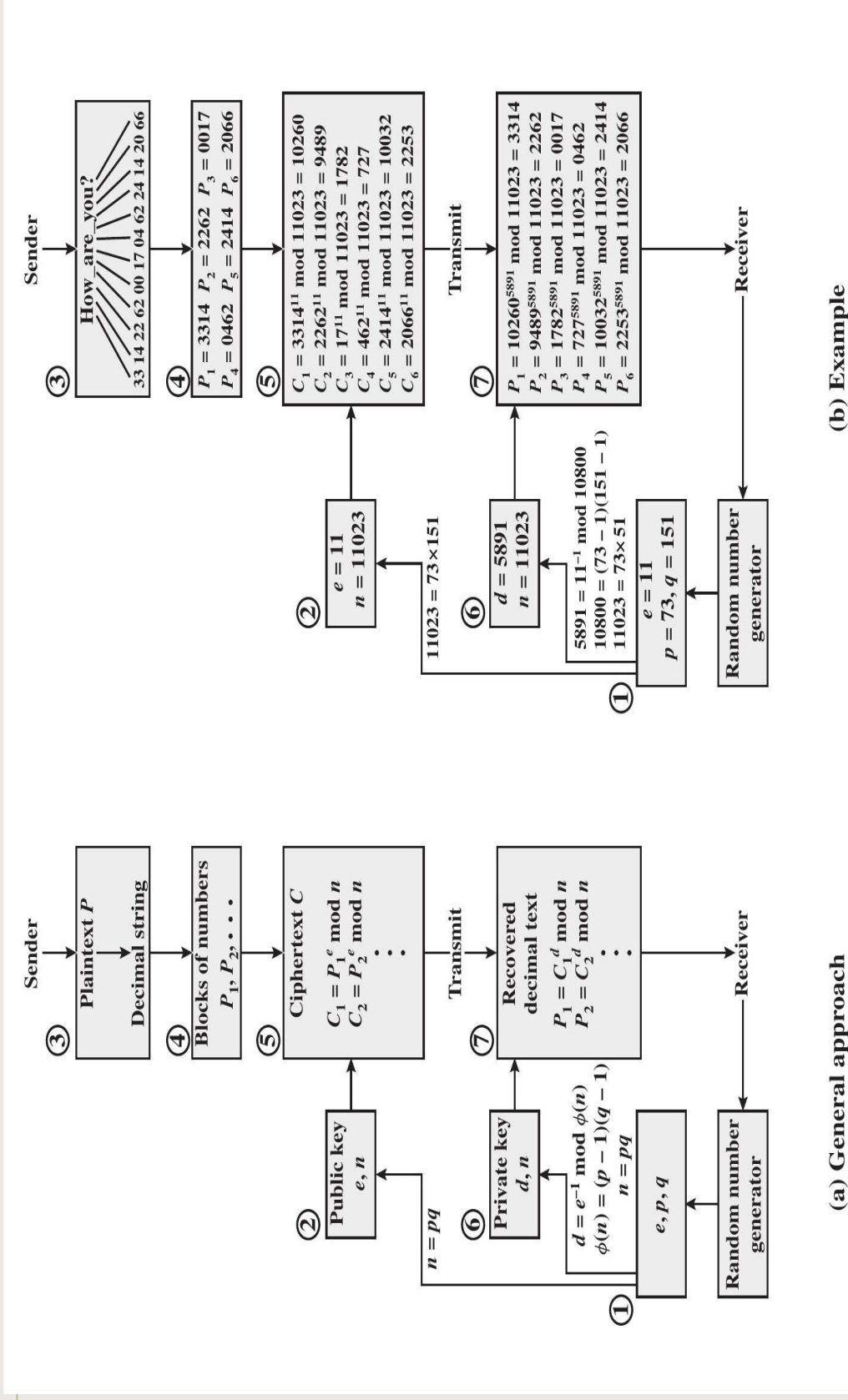
RSA ENCRYPTION... AN EXAMPLE

$$C = M^e \bmod n \quad M = C^d \bmod n$$

Public key: (5, 119) Private key: (77, 119)



RSA Processing of Multiple Blocks



RSA Exercise

- Perform encryption and decryption using RSA algorithm
 - 1) $p=5, q=11, e=3$ & $M=9$
 - 2) $p=11, q=13, e=11, M=7$
- In a public-key system using RSA, you intercept the ciphertext $C = 10$ sent to a user whose public key is $e = 5$, $n = 35$. What is the plaintext M ?

RSA Security

- Possible approaches to attacking RSA are:
 - Brute force key search - This involves trying all possible private keys but infeasible given size of numbers.
 - Mathematical attacks - based on difficulty of computing $\phi(n)$, by factoring modulus n
 - Timing attacks - on running time of decryption
 - Chosen ciphertext attacks - given properties of RSA

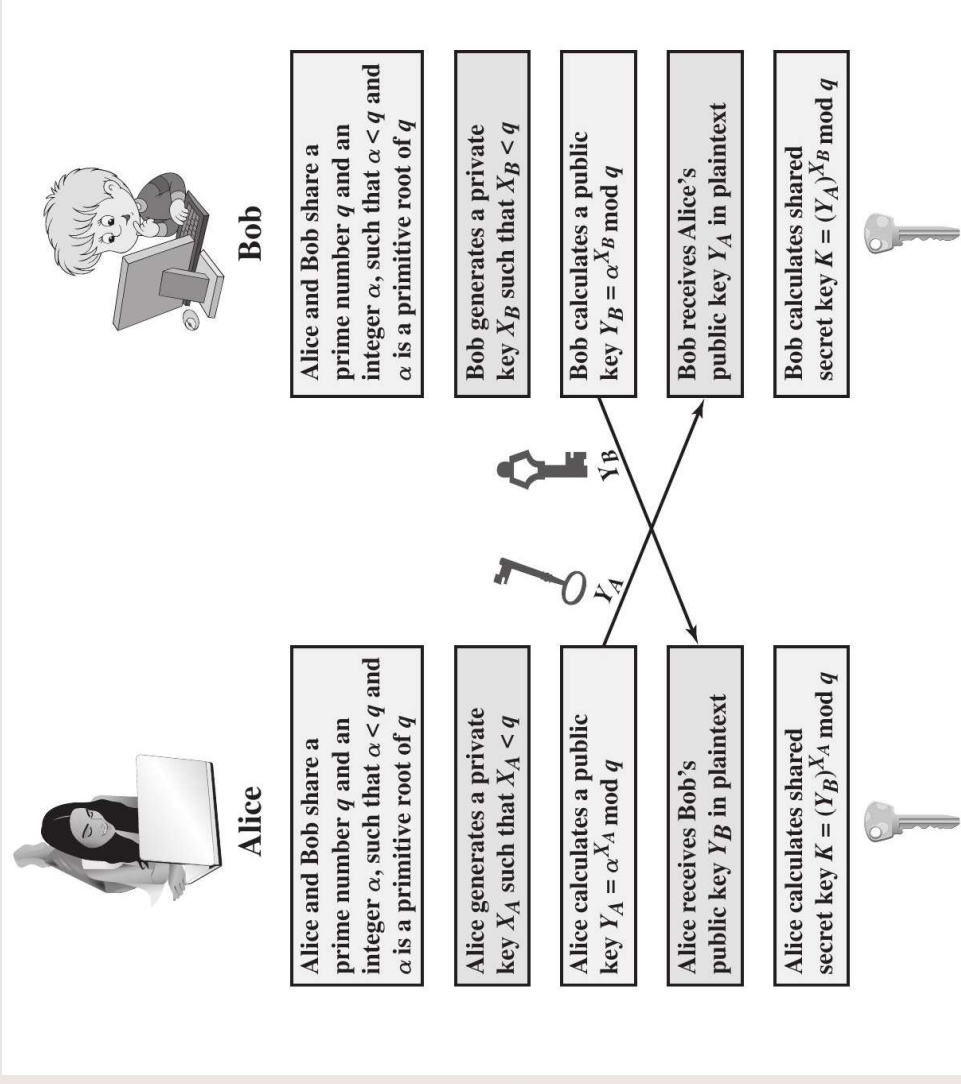
Diffie-Hellman Key Exchange Algorithm

- To enable two users to securely exchange a key that can then be used for subsequent symmetric encryption of messages.
- Security of Diffie-Hellman relies on the computational difficulty of the discrete log problem.

Discrete Logarithm

- If a is a primitive root of the prime number p , then the numbers $a \bmod p, a^2 \bmod p, \dots, a^{p-1} \bmod p$ are distinct and consist of the integers from 1 through $p - 1$ in some permutation.
- For any integer b
$$b \equiv a^i \pmod{p} \text{ where } 0 \leq i < (p - 1)$$
- The exponent i is referred to as the discrete logarithm $\text{dlog}_{a,p}(b)$ of b for the base $a, \bmod p$

Diffie-Hellman algorithm



- These two calculations produce identical results:

$$\begin{aligned} K &= (Y_B)^{X_A} \bmod q \\ &= (\alpha^{X_B} \bmod q)^{X_A} \bmod q \\ &= (\alpha^{X_B})^{X_A} \bmod q \\ &= \alpha^{X_B X_A} \bmod q \\ &= (\alpha^{X_A})^{X_B} \bmod q \\ &= (\alpha^{X_A} \bmod q)^{X_B} \bmod q \\ &= (Y_A)^{X_B} \bmod q \end{aligned}$$

- Two sides have exchanged a secret value
- This secret value is used as shared symmetric secret key

Security of the Diffie-Hellman

- Let an adversary observes the key exchange and know q , α , Y_A , and Y_B
- To determine the private key of user B, an adversary must compute

$$X_B = d \log_{\alpha, p}(b)$$

- the adversary can calculate K as

$$K = (Y_A)^{X_B \bmod q}$$

- It is relatively easy to calculate exponentials modulo a prime,
- it is very difficult to calculate discrete logarithms.
- For large primes, it is considered infeasible.

Example

- $q = 353$ and a primitive root of 353 , $\alpha = 3$, $X_A = 97$, $X_B = 233$

Public Keys:

A computes $Y_A = 3^{97} \bmod 353 = 40$.

B computes $Y_B = 3^{233} \bmod 353 = 248$

The Common Private Key:

A computes $K = (Y_B)^{X_A} \bmod 353 = 248^{97} \bmod 353 = 160$.

B computes $K = (Y_A)^{X_B} \bmod 353 = 40^{233} \bmod 353 = 160$.

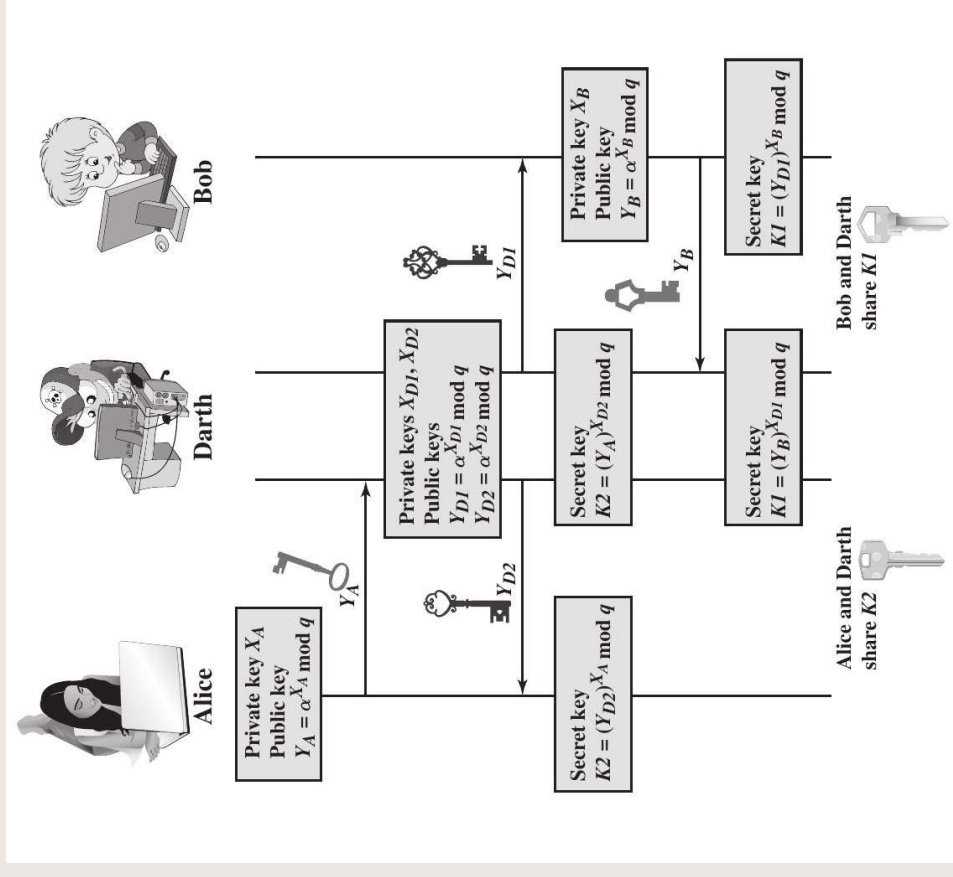
Brute Force Attack to determine the secret key 160.

- Let an attacker would have available the following information:
 $q = 353; \alpha = 3; Y_A = 40; Y_B = 248$
- Want to know the solution of $3^a \bmod 353 = 40$ or $3^b \bmod 353 = 248$
- Using brute-force approach is to calculate powers of 3 modulo 353, stopping when the result equals either 40 or 248.
- The desired answer is reached with the exponent value of 97, i.e. $3^{97} \bmod 353 = 40$.
- With larger numbers, the problem becomes impractical.

Man-in-the-Middle Attack

Suppose Alice and Bob wish to exchange keys, and Darth is the adversary.

1. Darth prepares for the attack by generating two random private keys X_{D1} and X_{D2} and then computing the corresponding public keys Y_{D1} and Y_{D2} .
2. Alice transmits Y_A to Bob.
3. Darth intercepts Y_A and transmits Y_{D1} to Bob. Darth also calculates $K2 = (Y_A)^{X_{D2}} \bmod q$.
4. Bob receives Y_{D1} and calculates $K1 = (Y_{D1})^X \bmod q$.
5. Bob transmits Y_B to Alice.
6. Darth intercepts Y_B and transmits Y_{D2} to Alice. Darth calculates $K1 = (Y_B)^{X_{D1}} \bmod q$.
7. Alice receives Y_{D2} and calculates $K2 = (Y_{D2})^{X_A} \bmod q$.
 - All future communication between Bob and Alice is compromised.
 - vulnerable to such an attack because it does not authenticate the participants. This vulnerability can be overcome with the use of digital signatures and public-key certificates



Elgamal Cryptosystem

- a prime number q and α , which is a primitive root of q . User A generates a private/public key pair as follows:

1. Generate a random integer X_A , such that $1 < X_A < q - 1$.
2. Compute $Y^A = \alpha^{X_A} \bmod q$.
3. A's private key is X_A and A's public key is $\{q, \alpha, Y_A\}$.

Any user B that has access to A's public key can encrypt a message as follows:

1. Represent the message as an integer M in the range $0 \leq M \leq q - 1$. Longer messages are sent as a sequence of blocks, with each block being an integer less than q .
2. Choose a random integer k such that $1 \leq k \leq q - 1$.
3. Compute a one-time key $K = (Y_A)^k \bmod q$.
4. Encrypt M as the pair of integers (C_1, C_2) where

$$C_1 = \alpha^k \bmod q; C_2 = KM \bmod q$$

User A recovers the plaintext as follows:

1. Recover the key by computing $K = (C_1)^{X_A} \bmod q$.
2. Compute $M = (C_2 K^{-1}) \bmod q$.

Elgamal Cryptosystem

- how K is recovered by the decryption process

$K = (Y_A)^k \bmod q$ K is defined during the encryption process

$K = (\alpha^{X_A} \bmod q)^k \bmod q$ substitute using $Y_A = \alpha^{X_A} \bmod q$

$K = \alpha^{kX_A} \bmod q$ by the rules of modular arithmetic

$K = (C_1)^{X_A} \bmod q$ substitute using $C_1 = \alpha^k \bmod q$

- using K , we recover the plaintext as

$$\begin{aligned} C_2 &= KM \bmod q \\ (C_2 K^{-1}) \bmod q &= KMK^{-1} \bmod q = M \bmod q = M \end{aligned}$$

Example

For example, let us start with the prime field $\text{GF}(19)$; that is, $q = 19$. It has primitive roots $\{2, 3, 10, 13, 14, 15\}$. We choose $\alpha = 10$.

Alice generates a key pair as follows:

1. Alice chooses $X_A = 5$.
2. Then $Y_A = \alpha^{X_A} \bmod q = 10^5 \bmod 19 = 3$
3. Alice's private key is 5 and Alice's public key is $\{q, \alpha, Y_A\} = \{19, 10, 3\}$.

Suppose Bob wants to send the message with the value $M = 17$. Then:

1. Bob chooses $k = 6$.
2. Then $K = (Y_A)^k \bmod q = 3^6 \bmod 19 = 729 \bmod 19 = 7$.

3. So

$$C_1 = \alpha^k \bmod q = 10^6 \bmod 19 = 11$$

$$C_2 = KM \bmod q = 7 \times 17 \bmod 19 = 119 \bmod 19 = 5$$

4. Bob sends the ciphertext $(11, 5)$.

For decryption:

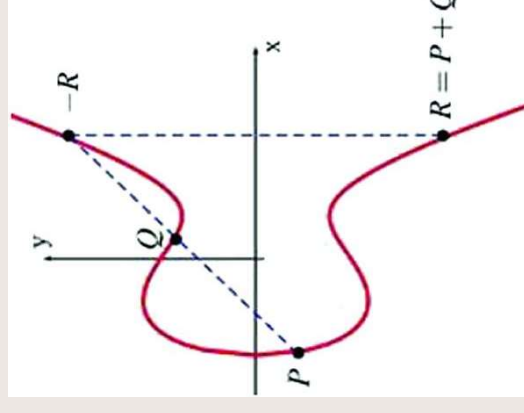
1. Alice calculates $K = (C_1)^{X_A} \bmod q = 11^5 \bmod 19 = 161051 \bmod 19 = 7$.
2. Then K^{-1} in $\text{GF}(19)$ is $7^{-1} \bmod 19 = 11$.
3. Finally, $M = (C_2 K^{-1}) \bmod q = 5 \times 11 \bmod 19 = 55 \bmod 19 = 17$.

Elliptic Curve Cryptography

- Elliptic-curve cryptography (ECC) is an approach to public-key cryptography based on the algebraic structure of elliptic curves over finite fields.
- An elliptic curve is a plane curve over a finite field which consists of the points satisfying the equation

$$y^2 = x^3 + ax + b,$$

- together with a special point at infinity, denoted $\square$.



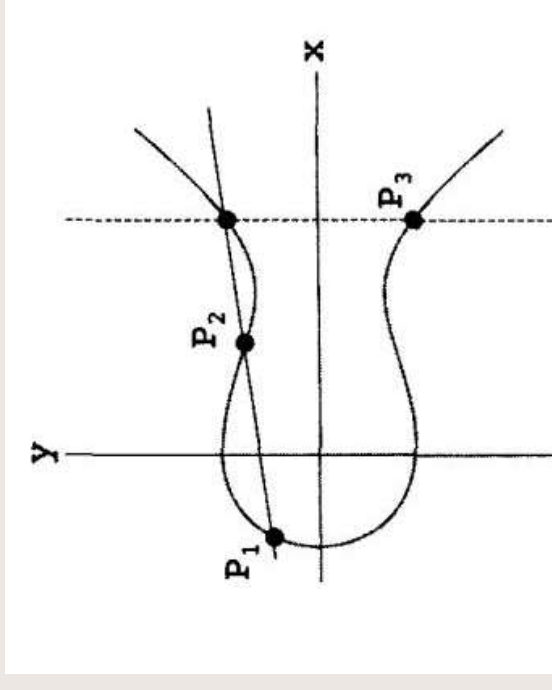
Elliptic Curve Cryptography

- The sum of the points P_1 and P_2 is defined as follows: First, a line is drawn through the two points. This line usually intersects the curve in one other point. If so, this intersection point is reflected about the x axis to obtain the point P_3 , which is defined to be the sum, that is,

$$P_3 = P_1 + P_2$$

- These points are calculated using *mod p*

$$y^2 = x^3 + ax + b \pmod{p}$$



Example

- consider the elliptic curve
- consider $x = 0, 1, 2, 3, 4$

$$\begin{aligned} x = 0 &\implies y^2 = 3 \implies \text{no solution mod } 5 \\ x = 1 &\implies y^2 = 6 = 1 \implies y = 1, 4 \text{ mod } 5 \\ x = 2 &\implies y^2 = 15 = 0 \implies y = 0 \text{ mod } 5 \\ x = 3 &\implies y^2 = 36 = 1 \implies y = 1, 4 \text{ mod } 5 \\ x = 4 &\implies y^2 = 75 = 0 \implies y = 0 \text{ mod } 5 \end{aligned}$$

- the points on the elliptic curve in equation are

$$(1, 1) \ (1, 4) \ (2, 0) \ (3, 1) \ (3, 4) \ (4, 0) \text{ and } \infty.$$

Given: curve E : $y^2 = x^3 + ax + b \pmod p$

$P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ on E

Find: $P_3 = (x_3, y_3) = P_1 + P_2$

Algorithm:

$x_3 = m^2 - x_1 - x_2 \pmod p$

$y_3 = m(x_1 - x_3) - y_1 \pmod p$

where $m = \begin{cases} (y_2 - y_1) \cdot (x_2 - x_1)^{-1} \pmod p & \text{if } P_1 \neq P_2 \\ (3x_1^2 + a) \cdot (2y_1)^{-1} \pmod p & \text{if } P_1 = P_2 \end{cases}$

Special case 1: If $m = \infty$ then $P_3 = \infty$

Special case 2: $\infty + P = P$ for all P

- to find the p oints $P_3 = (1, 4) + (3, 1)$

$$m = (1 - 4)/(3 - 1) = -3 \cdot 2^{-1} = -3 \cdot 3 = 1 \text{ mod } 5.$$

$$x_3 = 1^2 - 1 - 3 = -3 = 2 \text{ mod } 5$$

$$y_3 = 1(1 - 2) - 4 = -5 = 0 \text{ mod } 5.$$

- we have $(1, 4) + (3, 1) = (2, 0)$

ECC Diffie-Hellman

Global Public Elements	
$E_q(a, b)$	elliptic curve with parameters a, b , and q , where q is a prime or an integer of the form 2^m
G	point on elliptic curve whose order is large value n

User A Key Generation	
Select private n_A	$n_A < n$
Calculate public P_A	$P_A = n_A \times G$

User B Key Generation	
Select private n_B	$n_B < n$
Calculate public P_B	$P_B = n_B \times G$

Calculation of Secret Key by User A	
$K = n_A \times P_B$	

Calculation of Secret Key by User B	
$K = n_B \times P_A$	

- **Example** $y^2 = x^3 + ax + b$,
- take $p = 211$; $E_p(0, -4)$, which is equivalent to the curve $y^2 = x^3 - 4$; and $G = (2, 2)$.
- A's private key is $n_A = 121$,
- A's public key is $P_A = 121(2, 2) = (115, 48)$.
- B's private key is $n_B = 203$,
- B's public key is $203(2, 3) = (130, 203)$.
- The shared secret key is
- $121(130, 203) = 203(115, 48) = (161, 69)$.

Questions

- 1 Users A and B use the Diffie-Hellman key exchange technique with a common prime $q = 71$ and a primitive root $\alpha = 7$.
 - a. If user A has private key $X_A = 5$, what is A's public key Y_A ?
 - b. If user B has private key $X_B = 12$, what is B's public key Y_B ?
 - c. What is the shared secret key?
- 2 Consider a Diffie-Hellman scheme with a common prime $q = 11$ and a primitive root $\alpha = 2$.
 - a. Show that 2 is a primitive root of 11.
 - b. If user A has public key $Y_A = 9$, what is A's private key X_A ?
 - c. If user B has public key $Y_B = 3$, what is the secret key K shared with A?

Consider an Elgamal scheme with a common prime $q = 71$ and a primitive root $\alpha = 7$.

- a. If B has public key $Y_B = 3$ and A choose the random integer $k = 2$, what is the ciphertext of $M = 30$?
- b. If A now chooses a different value of k so that the encoding of $M = 30$ is $C = (59, C_2)$, what is the integer C_2 ?

Consider the elliptic curve

$$E : y^2 = x^3 + 7x + b \pmod{11}.$$

- a. Determine b so that the point $P = (4, 5)$ is on the curve E .
- b. Using the b found in part a, list all points on E .
- c. Using the b found in part a, find the sum $(4, 5) + (5, 4)$ on E .
- d. Using the b found in part a, find the point $3(4, 5)$.